What survives when a paper leaves the lab: re-testing 22 robot-learning papers on a €1,350 rig, and auditing our own pre-registration

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Abstract. Robot-learning papers are, with few exceptions, demonstrated in simulation or on hardware far more capable than what a small lab can afford, and it is seldom straightforward to tell from a paper which of its claims would survive the move to a cheaper rig. We re-tested 37 claims from 22 recent papers on an SO-101 arm driven by a Jetson Orin NX, a rig whose parts cost €1,350 and whose compute cost $45.10. For each claim we ran the cheapest baseline first and committed the pass mark to version control before the test. We find that 12 of the 37 claims hold on this hardware. The failures fall into three classes that we propose as a working taxonomy: methods that miss the real-time budget on the edge GPU, zero-shot judges that lose to a trivial signal, and latent planners that lose to the commanded joints. Because a register of this kind is only as good as the discipline behind it, we also audit our own pre-registration by reading the commit history of every gate. For most gates the bar provably precedes the result; for some the history cannot show the order, and we report those as well.

Code and rows: `github.com/ravediamond/paper-register-reproducibility` **Reproduce:** `make check`. Every number carries its source file and protocol commit.

1 The register

Our rig is an SO-101 arm and a Jetson Orin NX 16 GB. Most of the 22 papers in this study were demonstrated on setups far from it: 9 were shown in simulation and 7 on rigs above $20k, whereas only 2 used a setup under $5k, which is where our rig falls. The appendix table gives the rig of each paper, including the 3 in between and the 1 that does not say.

The register grew out of our own write-ups. Each re-test on this rig was published with its data, and there are 29 such write-ups so far. This paper collects them in one table and, in addition, audits how the table was kept.

To make the format concrete, consider one entry. An adaptive chunking paper claims that choosing the chunk length from the entropy of sampled chunks follows the phases of the task, long in transport and short at the grasp. On our edge GPU one such decision takes 4,751 ms, against a budget of 298 ms, so the entry records a fail on this rig; it says nothing about the method on the hardware it was shown on. Next to the verdict, the entry keeps the claim in the paper's own words, our pass mark, our measurement and the published rows.

For each claim we first run the cheapest signal that the learned model has to beat, on the same rows. For a predictor this is copy-last; for a latent it is the commanded joints; for a judge it is episode length. This ordering matters, because a learned method that cannot beat a trivial signal on the same data has not yet shown what it claims, whatever its absolute score.

We chose the papers for the stack this rig runs (world models, VLA policies, failure monitors, chunking, edge latency), so the failure rate describes this register and not the field. The register file lists every paper we have tested and every paper scheduled next, entered before its test.

2 Auditing our own pre-registration

A gate's pass mark had to be in version control before its result. Notes and commit dates can be edited by hand, but commit ancestry cannot be changed without rewriting history that others have already cloned, so ancestry is what we audit. For each gate we find the first commit that contains its bar and the first that contains its result, and compare them (`repro/prereg-check/scripts/prereg_audit.py`, abridged):

```
def ancestor(a, b):
    return subprocess.run(["git", "-C", str(RR), "merge-base", "--
     is-ancestor", a, b]).returncode == 0
...
            elif b[0] == res[0]:
                cls = "same-commit"
            elif ancestor(b[0], res[0]):
                cls = "bar-first"
            elif ancestor(res[0], b[0]):
                cls = "result-first"
```

Table 1 gives the classes for the 87 gates that ran and were logged by 2026-10-31. Most are bar-first and 0 are result-first. A same-commit gate is one where bar and result entered the repository together; it says nothing about which was written first. The log holds 2 further gates that are not in the table because we did not run them: one was superseded by a later test, and the other was dropped because neither of its outcomes would have changed what we built.

The first pass used text search alone and found 31 same-commit gates, an artefact of batch logging. For each of them we went back to the earlier file that holds its bar: 21 gates moved to bar-first and 4 to no-bar, because their

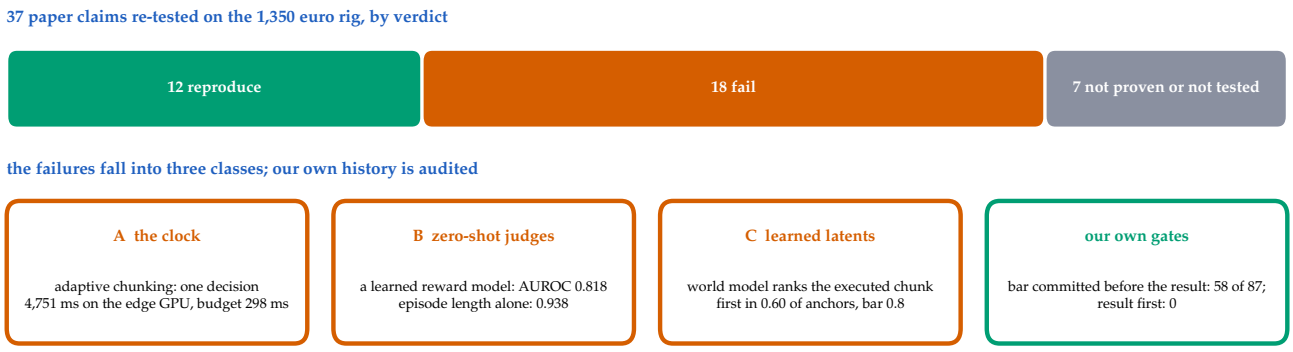


Figure 1: what survived when the papers left the lab: of the 37 paper claims, 12 reproduce and 18 fail; the failures fall into three classes (the clock, zero-shot judges against a trivial signal, learned latents against the commanded joints); our own gates have the bar committed before the result in 58 of 87 and the result first in 0.

Table 1: Commit-order class of each gate. The audited column is the one the paper uses; the first pass is what text search alone gave on the same gates.

class	first text pass	audited, 2026-10-31	audited, today
bar-first	36	58	61
same-commit	31	6	7
result-first	—	0	0
no bar	19	23	23
unresolved	1	—	—
gates	87	87	91

Table 2: The three gates whose verdict flips somewhere inside its 95 % interval.

gate	n	estimate	95 % CI	bar	point verdict
colour judge (no learning), accuracy	87	94.2 %	89.6–98.9 %	$\geq$ 95 %	fail
oscillation among failed episodes	25	24.0 %	8.0–40.0 %	$\geq$ 10 %	pass
judge threshold, false interventions	42	2.4 %	0.0–7.1 %	$\leq$ 5 %	pass

bar cell declares none, while 6 stayed same-commit. The audit table in the repository gives the evidence line for every move. We did not re-audit the gates that were already bar-first. One case is still open: the reward-model progress test of 2026-11-01, which text search calls same-commit although its protocol (`6241570`) is older than the log commit.

The early gates mixed point estimates with intervals of different kinds. We therefore recomputed 10 of them from stored rows with one statistic, the episode bootstrap (2,000 resamples, percentile 95 % CI). For 3 of them the bar falls inside the interval (Table 2). The colour judge is a fail on the point estimate, but its interval does not settle the question; the false-intervention threshold was fitted to meet its rate, so that interval is descriptive only. The register keeps the pre-registered point verdicts and marks these three as fragile.

All 29 write-ups rebuild from their published rows. The check for the cold reruns prints:

```
$ python3 repro/cold-reproduction/l0_check.py
ok  posts: 29 (report 29)
```

Table 3: Verdicts by paper and by claim row (claim rows include our own baseline comparisons).

verdict	papers	claim rows
reproduces	4	17
fails	13	20
not proven	3	3
not tested	1	4
withdrawn	1	1
total	22	45

```
ok  L0 PASS: 29 (report 29)
ok  L1 REPRO: 6 (report 6)
ok  L1 DIFF: 0 (report 0)
ok  L1 classes: {'REPRO': 6, -'': 3, 'SKIP': 8, 'FAIL-ENV': 12}
L0 PASS
```

We ran the cold reruns from a fresh clone with no credentials. Of the gates, 6 reran and matched, 5 of them exactly, and 12 more stop at the world-model environment, which is not yet published; publishing it would unblock most of them. The remaining 11 require a larger GPU, a private checkpoint or the arm, or depend on a step the write-up leaves out. We also checked the register's citations and paper-side fields against the papers' PDFs and corrected them. The register names 38 arXiv ids; the ids missing from our papers mirror all belong to scheduled tests, and one tested paper has no arXiv id, so it could not be checked against a PDF.

3 Results

Table 3 counts the verdicts by paper and by claim row.

Finding. On claims taken from papers, 12 of 37 reproduce and 18 fail. The remaining 8 rows test our own baselines, for example that the commanded joints beat the world model, and 5 of those hold.

4 Three failure classes

The failures we observed are not of one kind, and we found it useful to distinguish three. The adaptive chunking entry of §1 misses the real-time budget on the edge GPU; the method may well work on faster hardware, and the failure is one of the clock. We call this class A.

In the second kind of failure, a model judges outcomes without having been trained on this rig. We ran a learned reward model as a failure detector and it reached an AUROC of 0.818, whereas episode length alone reaches 0.938 on the same rollouts. Judges of success, failure or progress that lose to a trivial signal in this way form class B. The third kind concerns learned latents. A world model ranked the executed chunk first in 0.60 of anchors against a pass mark of 0.8, while an action-only baseline reaches 0.68–0.90. Latents used to plan, rank or track progress that lose to the joints, to copy-last or to an action-only baseline form class C.

These examples illustrate the classes and do not count them. The classes are our proposal, drawn from the author's reading of the 20 failed rows, and we expect them to be refined as the register grows.

5 Limitations

The main confound in this study is observer bias: one operator, the author, ran the rig and labelled the outcomes, and labels were given at rest and not always blind to the arm. Where commit order shows it, the pass mark predates the run, so a label could not have moved the bar. For 29 of the 87 gates, however, the repository cannot show whether bar or result came first, and 21 of the bar-first verdicts rest on matching a bar text in an earlier file. Under the bootstrap, 3 verdicts are fragile. Cold reruns cover 6 gates so far, with the unpublished world-model environment blocking most of the rest.

6 Related work

Several works evaluate policies on the same arm. A standardized real-world benchmark reports a taxonomy of failures and recoveries for VLA policies [13], and a fleet of SO-101 cells evaluates policies at scale [14]. The vla-eval tool runs VLAs across 14 simulation benchmarks [8], and a benchmark of demonstration quality studies one method family [22]. Our work differs in its object: we test the claims behind methods (judges, world models, latency) against thresholds fixed before the run, and we audit our own pre-registration. The closest work in spirit is RoboSeed [29], which scores VLA and world-action models on five dimensions, including reproduction evidence and deployment affordability. Its reproduction-evidence score counts five audit milestones (configuration mapping, task records, task coverage, training records and seeds), and its authors are careful to state that the score is not a probability that a result is correct. In contrast, we give one pass or fail per claim on a €1,350 rig. Concerns of this kind are not new to machine learning: the reproducibility of deep reinforcement learning results [28] and the statistical fragility of point estimates reported from a handful of runs [27] have been examined before, and our register can be read as the robot-learning counterpart of that line of work, with the added constraint that the hardware itself is part of what is being reproduced.

7 Conclusion

Conclusion. We re-tested 37 paper claims on a rig that cost €1,350 in parts and $45.10 in compute, and 12 of them held. Every row behind the verdicts is published, and every number in this paper can be rebuilt from the repository (`github.com/ravediamond/paper-register-reproducibility`). For 29 of 87 gates our own history cannot show the order of bar and result. Our findings suggest that re-testing a paper on a cheap rig should start with its cheapest baseline on the same rows, with the threshold committed before the run; we hope that the register and its audit offer a template for doing so.

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Appendix: the register

paper	their rig	claim tested	bar	our result	verdict
LeWorldModel [9]	sim	A small JEPA world model learns the arm's dynamics well enough to plan with it.	Dynamics: beat copy-last, every corruption ≥ 1.3 at $K = 30$. Planning: rank the executed chunk first among 8 in ≥ 0.8 of anchors.	dynamics 0.31 vs 0.89 copy-last (reproduces); planning top-1 0.60 vs 0.68–0.90 for an action-only baseline (fail)	fail
DINO-WM [1]	sim	A predictor on frozen DINOv2 patches plans better than one on a learned latent.	G1b top-1 ≥ 0.8 at some target distance.	arm-pose R^2 0.92 (vs 0.57 for W1); G1b top-1 0.55–0.61; no-model baseline 0.68–0.90	fail
RoboJEPA [20]	>$20k	Pretrained RoboJEPA predicts our latents, uses the actions, and runs on the robot's computer.	9-step L1 ≥ 15 % below copy-last (300M) and ≥ 5 % below shuffled actions; 2-view encoder ≤ 200 ms on the Orin.	11.8 % below copy-last (300M; 1B 15.8 % on 30 clips); 9–10 % below shuffled; encoder 526 ms on the Orin NX	fail
H-JEPA [19]	sim	A slow second level makes latent distance track task progress.	Progress monotonicity Spearman $\rho \leq -0.80$; false stalls ≤ 10 %.	level-2 ρ -0.73; false stalls 52 %	fail
D-JEPA [18]	<$5k	A decision-aligned ranking operator on a frozen world model picks better action chunks.	G1b regret ≤ 0.10 and $\leq$ the action-only baseline + 0.05.	regret 0.263 (W1 alone 0.263; action-only -0.007)	not tested
ContactGuard [16]	$5–20k	An event-anchored probe on imagined latents predicts grasp failure better than the current latent.	AUROC ≥ 0.85 and ≥ 0.05 above the current latent.	0.70 imagined vs 0.82 current	fail
FAIL-Detect and SAFE [2] [5]	>$20k	A conformal band on a learned failure score keeps false alarms at α while catching failures.	Successes flagged $\leq 1/20$; failures flagged ≥ 50 %.	22.7 % of successes flagged at $\alpha = 5$ %; 46 % of failures caught	fail
ReAP [21]	$5–20k	A risk head on an action-conditioned latent warns before failures.	$\geq 5/8$ failures caught at $\leq 1/20$ false interventions; action-conditioned iff true $\geq$ shuffled + 0.05.	v1 AUROC 0.93 came from time and length; v2 0.789 vs 0.708 shuffled, 3/8 at 5/42	fail
VLA-Corrector [17]	sim	A verifier on observable history flags a failing rollout early enough to correct it.	$\geq 5/8$ failures at $\leq 1/20$ false interventions.	the paper's verifier family on our rollouts (a GRU on pooled RGB statistics and actions): 2/8 caught at 3/42 false interventions, AUROC 0.59; cosine drift of the W1 latent 0/8, AUROC 0.30	fail
Robometer [7]	>$20k	A 4B reward model judges SO-101 rollouts zero-shot.	Pooled AUROC ≥ 0.85 and within-policy ≥ 0.8 on ≥ 5 of 7 policies.	pooled AUROC 0.818 on 2099 ArmnetBench rollouts; episode length alone 0.938	fail
SmolVLA [3]	<$5k	A 450M VLA succeeds on an SO-101 task and can run on low-cost hardware.	10-trial operator-judged gate (no numeric pass mark registered); chunk inference within the async budget (667 ms).	9/10 on bowls; 974 ms in PyTorch, 238 ms with vla.cpp on the Orin NX	reproduces

Table 4: The register, one row per paper, generated from the register file (internal model and gate names kept). The rig price of €1,350 counts the parts with a stated price; 2 small parts are not priced yet. Read the verdict column against the rig each paper was shown on.

paper	their rig	claim tested	bar	our result	verdict
TurboVLA [15]	sim	Our claim that the architecture made the policy ignore its instruction.	Task swap $\geq 5°$ and per-layout ≥ 75 % $\rightarrow$ our port was the bug; $< 1°$ with the control $\geq 0.095 \rightarrow$ the data.	upstream's own trainer on our data: task swap 0.020°, per-layout 38.9 %; control, their released LIBERO checkpoint run here: 0.105 ($\approx 5.5°$ of pan)	withdrawn
VLA-JEPA [6]	sim	VLA-JEPA trains cheaply on SO-101 data and runs on the robot.	10-trial operator-judged gate (no numeric pass mark registered); for reranking (H0): oracle / mean error $\leq$ 0.7 and N = 8 $\leq 1.5\times$ the latency.	9/10 for $9.7; 355 ms per chunk on the Orin NX; reranking no-go (0.76, 1.84×)	reproduces
REKI [23]	sim	Policies without memory oscillate between poses at a measurable rate.	≥ 10 % of failed episodes oscillate (their criterion).	20.3 % of all episodes, 24.0 % of failures; REKI trained on my data: 19/20 + 19/20 on the arm, oscillates in 1/40	reproduces
Real-Time Chunking [4]	unstated	Conditioning a new chunk on the executing one removes the jump at chunk boundaries.	Best gradient-free method cuts the median jump ≥ 50 % against naive async, drift $\leq +10$ %, within 667 ms.	prefix overwrite cuts the jump 71–73 % against naive async at +0.4 ms; tuned LeRobot identity mode is close	not proven
HWM (hierarchical latent world models) [10]	>$20k	A high-level world model over learned macro-actions beats the low-level model beyond about 1 s (their Fig. 6 crossover).	Crossover in [1.0, 2.0] s; at 2 s high-level error $\leq 0.95 \times$ low-level; learned macro-actions $\geq$ net joint displacement on cosine.	no crossover: W1 $0.382 \pm 0.005 \rightarrow 0.563 \pm 0.007$, high level $0.769 \pm 0.009 \rightarrow 0.759 \pm 0.009$ (0.73–2 s); cosine 0.608 ± 0.007 vs 0.551 ± 0.006	not proven
LeJEPA identifiability [12]	sim	A SIGReg JEPA trained on goal-directed data recovers the true latent state only partially and unevenly.	W1's mean R^2 from latent to joints < 0.80; anisotropy (max $-$ min per-joint R^2) ≥ 0.20; the higher-R^2 model has the higher G1b top-1.	W1 mean R^2 0.565, anisotropy 0.639 (shoulder lift 0.91, wrist flex 0.27); frozen DINOv2-S 0.867 / 0.301	reproduces
SURVEYOR	sim	A threshold on the imagined-vs-reached latent distance accepts good plans and rejects bad ones ($\tau = 0.20$).	τ below the 10th percentile of non-equivalent pairs and ≤ 0.40; as a monitor $\geq 5/8$ failures at $\leq 2/42$; ≥ 0.05 AUROC over a shuffled drafter.	$\tau = 1.249$ on W1 latents of 69 real rollouts; as a monitor 3/8 at 6/42 (S = 30), AUROC 0.756 vs shuffled 0.560	fail
Adaptive Action Chunking (AAC) [11]	>$20k	Chunk length chosen from the entropy of N sampled chunks follows the task's phases (long in transport, short at the grasp), lowers the boundary jump, and costs little with batched sampling.	median h* in transport $\geq$ grasp window + 2; jump with AAC + overwrite $\leq$ fixed + overwrite at d = 8; an N = 20 decision ≤ 298 ms on the Orin.	h* 2 in transport and 2 at the grasp (86 % of decisions at the floor); jump 0.973 vs 0.940; N = 20 decision 4,751 ms (sequential) on the Orin	fail
PACT-VLA [24]	$5–20k	Picking among N sampled chunks by their flow residual under the pre-commit observation fits what happens next better than a random or stale pick, with no added control-step delay.	PACT beats a random pick and the stale-observation residual at d = 8 (CIs < 0); its delay on the Orin $\leq$ the naive delay; at that delay it still beats a single sample.	offline $-0.27°$ vs random and vs the stale residual (both CIs < 0); on the Orin 1,200 ms per cycle, 37 control ticks vs 8	fail
Horizon × depth (Euler steps vs execution horizon) [25]	>$20k	With long open-loop commitment, cutting a flow-matching VLA's Euler steps from 10 to 2 contracts the action cloud around an unchanged centroid, saves most of the inference time and avoids edge-GPU thermal throttling.	median $R(2)/R(10) \leq 0.70$ with the centroid within 0.05 relative and 3°; N = 2 latency $\leq 0.85 \times$ N = 10 on the Orin, N = 10 within 5 % of 238 ms.	R(2)/R(10) 0.78 (CI 0.73–0.81), centroid 0.018 and 2.35°; 196.1 vs 237.5 ms per chunk (0.83, vision 151 ms); no throttling at N = 10 on the Orin, at MAXN_SUPER or 25 W	not proven
REMORA [26]	>$20k	Reading a progress distribution out at the supported peak nearest the previous estimate steadies the estimate and lets it fall during real setbacks, with no worse detection of the stages.	Against per-frame argmax: premature not higher and missed $\leq$ +2 points (P1); flips halved, CI upper ≤ 0.7 (P2); a decrease inside ≥ 50 % of the reversal windows with ≤ 20 % false regressions (P3).	premature 0.22 vs 0.25, missed equal (P1 pass); flips $0.70 \times$ argmax, CI 0.48–0.94 (P2 fail); setback recall 2/7, false 0.52 (P3 fail); MAE 0.184 vs median 0.218	fail

Table 4, continued.

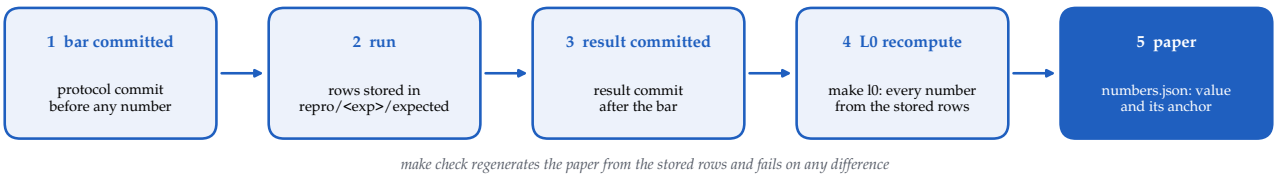


Figure 2: how every number in this paper is made: a protocol with its bar is committed before any number exists (1), the run stores its rows under repro/ (2), the result is committed after the bar and the commit order is audited (3), make l0 recomputes every number from the stored rows (4), make_paper.py writes numbers.json with each value's anchor and fills the text from it (5); make check regenerates the paper and fails on any difference.